

U.S. Futures and Options Package

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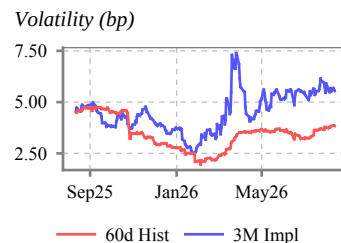
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Short-Dated SOFR Swaption Volatility Report

Options on 1yr Swaps

Mat	Implied Vol.	5d	20d
%	bp	Chg	Chg
1m	18.58	4.74	-0.01
3m	20.70	5.35	0.05
6m	22.00	5.74	-0.11
1y	23.03	5.97	-0.15

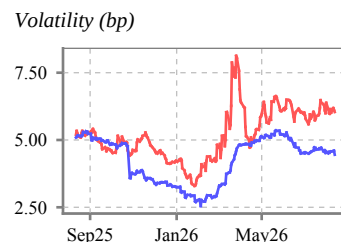
Historical BP Vol.	10d	20d	60d
	3.38	3.58	3.84
*1dc	1wc	2wc	
	3.58	4.05	2.82



Options on 2yr Swaps

Mat	Implied Vol.	5d	20d
%	bp	Chg	Chg
1m	20.96	5.39	0.01
3m	22.13	5.72	0.06
6m	22.54	5.84	-0.10
1y	22.92	5.93	-0.15

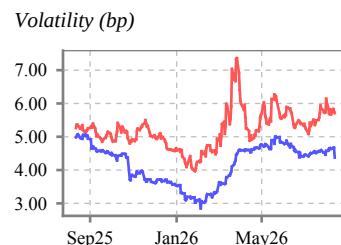
Historical BP Vol.	10d	20d	60d
	4.42	4.07	4.45
*1dc	1wc	2wc	
	4.07	4.00	2.87



Options on 5yr Swaps

Mat	Implied Vol.	5d	20d
%	bp	Chg	Chg
1m	19.86	5.15	0.00
3m	20.51	5.33	0.03
6m	20.62	5.38	-0.11
1y	20.87	5.47	-0.10

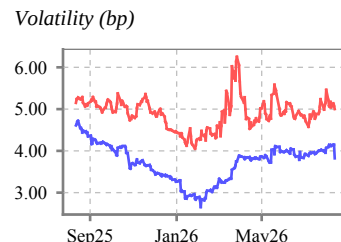
Historical BP Vol.	10d	20d	60d
	5.01	4.24	4.37
*1dc	1wc	2wc	
	4.24	3.20	2.35



Options on 10yr Swaps

Mat	Implied Vol.	5d	20d
%	bp	Chg	Chg
1m	16.00	4.33	-0.13
3m	16.89	4.58	-0.06
6m	17.60	4.80	-0.10
1y	18.39	5.05	-0.08

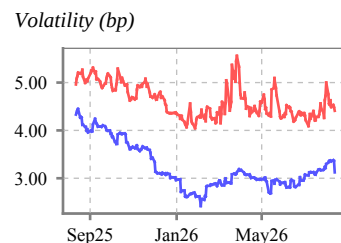
Historical BP Vol.	10d	20d	60d
	4.54	3.84	3.82
*1dc	1wc	2wc	
	3.84	2.41	1.78



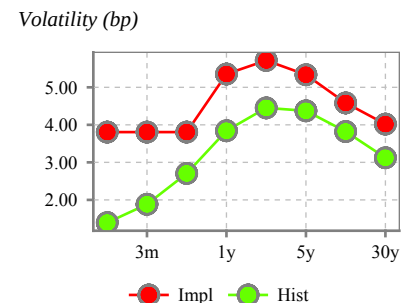
Options on 30yr Swaps

Mat	Implied Vol.	5d	20d
%	bp	Chg	Chg
1m	13.38	3.81	-0.20
3m	14.10	4.02	-0.13
6m	14.97	4.27	-0.13
1y	16.11	4.60	-0.10

Historical BP Vol.	10d	20d	60d
	3.69	3.27	3.12
*1dc	1wc	2wc	
	3.27	2.06	1.43

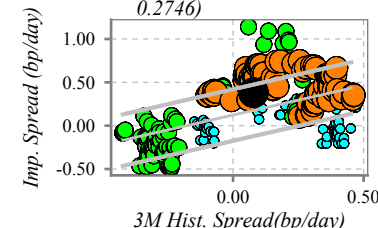


Maturity Structure of 3M Vol.



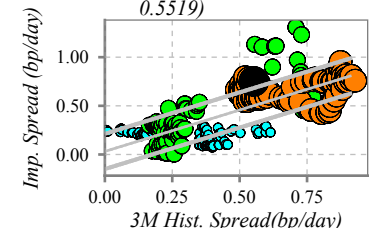
3Mx2Y - 3Mx5Y

$$Y = 0.6874 X1 + 0.1227 \quad (\text{adjR2} = 0.2746)$$

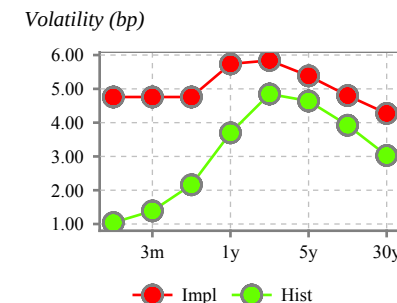


3Mx5Y - 3Mx10Y

$$Y = 0.8512 X1 + 0.0314 \quad (\text{adjR2} = 0.5519)$$

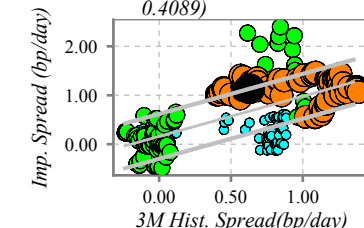


Maturity Structure of 6M Vol.



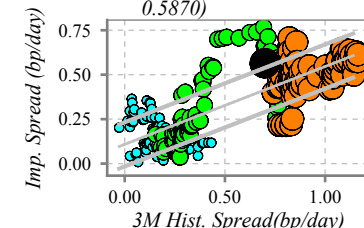
3Mx2Y - 3Mx10Y

$$Y = 0.8190 X1 + 0.1603 \quad (\text{adjR2} = 0.4089)$$



3Mx10Y - 3Mx30Y

$$Y = 0.4376 X1 + 0.1083 \quad (\text{adjR2} = 0.5870)$$



	Hist Vol.		Implied Vol.	
Mat/Und	hold	curr	hold	curr
3mx01y	3.70	3.84	4.77	5.35
3mx02y	4.82	4.45	5.48	5.72
3mx05y	5.43	4.37	5.70	5.33
3mx10y	5.36	3.82	5.53	4.58
3mx30y	4.96	3.12	5.15	4.02

1. Historical BP Vol. is calculated the 10-day, 20-day, and 60-day standard deviations of daily spot SOFR rate changes on the tenor specified in the table
2. Regression of difference in 3M implieds vs difference in 3M historicals. Regression over past 12m.
3. *One-month realized BP Vol. over 1-day, 1-week and 2-week change.

Derivatives Strategy

	Treasury Bond					Treasury Note				5 Year Treasury Note			
	Sep 26	Oct 26	Nov 26	Dec 26	Jan 27	Sep 26	Oct 26	Nov 26	Dec 26	Sep 26	Oct 26	Nov 26	Dec 26
VOLATILITY													
Implied (%)													
Current	8.70	9.28	9.41	9.67	9.74	4.25	4.77	4.81	4.92	3.03	3.41	3.42	3.52
Change (1d)	-0.84	-0.34	-0.20	-0.24	-0.20	-0.66	-0.32	-0.27	-0.24	-0.57	-0.17	-0.10	-0.09
20 day High	11.30	10.98	10.81	10.84	9.94	5.76	5.77	5.64	5.70	3.90	3.91	3.83	3.88
20 day Low	7.89	8.26	8.52	8.87	8.63	4.25	4.59	4.73	4.86	3.03	3.29	3.36	3.48
Business Day	8.18	9.17	9.27	9.52	9.64	3.99	4.71	4.73	4.85	2.85	3.37	3.37	3.46
Historical (%)													
10 day	9.23	9.41	9.41	9.41	9.44	4.79	4.91	4.91	4.91	3.22	3.36	3.36	3.36
20 day	7.95	8.07	8.07	8.07	8.06	4.14	4.24	4.24	4.24	2.79	2.87	2.87	2.87
50 day	7.34	7.36	7.36	7.36	6.40	4.10	4.14	4.14	4.14	2.88	2.91	2.91	2.91
2Yr Event Risk	-	-	-	-	-	-	-	-	-	-	-	-	-
Volatility Cones (%)													
3 Year High	23.31	17.75	16.64	15.86	15.19	12.23	9.05	8.59	7.99	8.71	6.06	5.75	5.52
3 Year Average	9.84	10.25	10.36	10.41	10.46	5.25	5.46	5.54	5.58	3.55	3.69	3.75	3.79
3 Year Low	2.95	5.59	6.02	6.26	6.40	1.93	2.73	3.05	3.17	1.14	1.71	2.01	2.11
% Historic Above Implied	57.18	62.19	64.30	64.56	60.87	65.22	72.73	71.81	68.51	59.82	62.85	61.92	54.94
BASIS POINT VOLATILITY													
Implied (bp)													
Current	-	-	-	-	-	4.6	5.1	5.1	5.3	4.9	5.5	5.5	5.7
Change (1d)	-	-	-	-	-	-0.7	-0.3	-0.3	-0.2	-0.9	-0.3	-0.2	-0.1
Change (5d)	-	-	-	-	-	-0.7	-0.1	-0.1	-0.2	-0.8	0.0	0.0	0.0
Change (20d)	-	-	-	-	-	-0.2	0.1	0.1	0.1	-0.4	0.2	0.1	0.1
20 day High	5.4	5.2	5.1	5.2	4.8	6.2	6.1	6.0	6.1	6.3	6.3	6.2	6.3
20 day Low	3.9	4.0	4.2	4.4	4.2	4.6	4.9	5.0	5.2	4.9	5.3	5.4	5.6
Historical (bp)													
10 day	4.8	4.6	4.6	4.6	5.7	5.1	5.2	5.2	5.2	5.2	5.1	5.1	5.1
20 day	5.1	4.9	4.9	4.9	5.2	4.4	4.4	4.4	4.4	4.5	4.4	4.4	4.4
50 day	4.7	4.4	4.4	4.4	4.0	4.3	4.3	4.3	4.3	4.5	4.5	4.5	4.5
Volatility Cones (bp)													
3 Year High	13.9	11.2	10.2	9.7	9.2	13.5	9.7	9.2	8.4	13.7	9.8	9.2	8.6
3 Year Average	6.0	6.2	6.4	6.4	6.5	5.6	5.8	5.9	5.9	5.6	5.9	5.9	6.0
3 Year Low	1.2	3.0	3.3	3.6	4.0	2.1	2.9	3.2	3.4	1.8	2.7	3.1	3.3
% Historic Above Implied	0.0	0.0	0.0	0.0	0.0	61.0	69.7	69.2	67.6	55.5	54.0	56.9	53.0
Yield Change Required to... (bp)													
Offset 1 C-day Decay	3.0	2.8	2.9	2.9	2.8	3.5	3.9	4.0	4.0	3.8	4.0	4.0	4.0
Offset 1 B-day Decay	3.6	3.4	3.5	3.5	3.4	4.2	4.7	4.8	4.8	4.6	4.8	4.8	4.9
Offset 1 Month Decay	-	15.1	15.9	15.9	15.3	-	20.7	21.6	22.0	-	20.1	21.3	21.8
Break Even at Expiry	7.2	15.2	19.8	23.1	25.8	8.5	20.9	27.1	32.2	9.4	21.0	27.0	32.1
BPV per Contract	\$161.52	\$167.35	\$167.35	\$167.35	\$173.57	\$68.26	\$67.09	\$67.09	\$67.09	\$42.84	\$45.60	\$45.60	\$45.60
Cal/Bus Days To Expiry	9/7	44/32	72/52	100/72	134/96	9/7	44/32	72/52	100/72	9/7	44/32	72/52	100/72
Futures Price	109-01	108-17	108-17	108-17	108-06	108-17+	108-09	108-09	108-09	106-11+	106-08	106-08	106-08

VOLATILITY is price volatility for bond, note and index options; yield volatility for money market options

Volatility Cones are based on number of days to expiration. For example, if 40 days remain, we look at actual volatility of the nearby contract in every 40 day period over the last 2 years

% Historical Above Implied shows how often actual volatility exceeded current implied volatility. A higher number means cheaper options.

BASIS POINT VOLATILITY is basis point yield volatility for all options.

Yield Change Required to... is based on an at-the-money straddle. Yield change for bond, note and money market options; Price change for equity and commodity options

* Historical volatility is taken from old otr bond This

Derivatives Strategy

	Secured ON Financing Rate 3M													
	Aug 26	Sep 26	Oct 26	Nov 26	Dec 26	Jan 27	Mar 27	Jun 27	Sep 27	Dec 27	Mar 28	Jun 28	Sep 28	Dec 28
VOLATILITY Implied (%)														
Current	12.19	11.21	17.54	17.23	16.80	20.70	19.90	21.63	22.78	23.51	24.30	24.59	24.55	24.36
Change (1d)	-9.45	-1.88	-1.36	-1.14	-1.28	-0.80	-0.89	-0.77	-0.39	-0.53	-0.33	-0.21	-0.20	-0.43
20 day High	22.62	18.63	21.36	20.29	19.25	22.58	21.82	23.35	24.04	24.70	25.36	25.42	25.75	25.56
20 day Low	10.70	11.21	17.54	17.23	16.80	20.70	19.90	21.63	22.78	23.51	24.30	24.59	24.55	24.36
Business Day	10.11	11.11	17.29	16.95	16.62	20.62	19.87	21.59	22.77	23.46	24.27	24.54	24.52	24.31
Historical (%)														
10 day	9.42	9.42	14.03	14.03	14.03	18.11	18.11	20.33	21.24	21.26	21.27	21.06	21.16	21.31
20 day	13.50	13.50	14.93	14.93	14.93	16.99	16.99	18.35	18.89	18.38	18.06	17.82	17.80	17.98
50 day	15.08	15.08	18.61	18.61	18.61	20.49	20.49	20.82	20.47	20.14	19.64	19.48	19.03	18.66
2Yr Event Risk	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Volatility Cones (%)														
3 Year High	145.13	60.85	46.94	42.02	40.16	36.48	36.20	33.91	32.25	31.00	29.58	31.73	32.33	32.00
3 Year Average	22.07	27.69	28.94	29.31	29.43	29.42	29.38	28.79	28.28	27.82	27.11	26.75	26.87	27.92
3 Year Low	0.00	10.43	15.13	16.18	17.53	18.69	20.81	23.53	23.67	23.93	24.92	24.15	23.82	23.78
% Historic Above Implied	63.50	99.47	95.26	96.71	97.76	89.86	89.72	81.42	73.25	64.82	56.65	42.03	33.73	25.16
BASIS POINT VOLATILITY Implied (bp)														
Current	2.9	2.7	4.4	4.3	4.2	5.3	5.1	5.7	6.0	6.0	6.3	6.3	6.4	6.4
Change (1d)	-2.3	-0.5	-0.4	-0.3	-0.3	-0.2	-0.2	-0.2	-0.1	-0.3	-0.1	-0.1	-0.1	-0.1
Change (5d)	-0.7	-0.4	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.1	-0.2	-0.2	-0.2	-0.1	-0.2
Change (20d)	0.1	-0.3	-0.2	-0.2	-0.2	0.0	-0.1	-0.1	0.0	-0.1	0.1	0.1	0.1	0.2
20 day High	5.7	4.7	5.6	5.4	5.1	6.1	5.8	6.3	6.5	6.5	6.7	6.7	6.7	6.7
20 day Low	2.6	2.7	4.4	4.3	4.2	5.3	5.1	5.7	6.0	6.0	6.2	6.2	6.3	6.2
Historical (bp)														
10 day	2.3	2.3	3.6	3.6	3.6	4.7	4.7	5.3	5.5	5.5	5.4	5.4	5.4	5.4
20 day	3.3	3.3	3.8	3.8	3.8	4.4	4.4	4.8	4.9	4.7	4.6	4.5	4.5	4.6
50 day	3.7	3.7	4.7	4.7	4.7	5.3	5.3	5.4	5.2	5.1	4.9	4.8	4.7	4.6
Volatility Cones (bp)														
3 Year High	1.9	0.8	0.6	0.6	0.6	0.5	0.5	0.5	0.5	0.5	0.4	0.5	0.5	0.5
3 Year Average	0.3	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
3 Year Low	0.0	0.1	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.4	0.3	0.3	0.3
% Historic Above Implied	60.2	96.4	85.8	88.5	88.3	79.6	77.5	63.4	52.3	38.7	25.4	17.7	14.2	19.4
Yield Change Required to... (bp)														
Offset 1 C-day Decay	2.2	2.2	3.6	3.5	3.5	4.3	4.2	4.5	4.7	4.8	4.9	4.9	4.9	4.8
Offset 1 B-day Decay	2.7	2.7	4.4	4.3	4.2	5.2	5.0	5.5	5.7	5.7	5.9	5.9	5.9	5.8
Offset 1 Month Decay	-	-	18.6	18.8	18.6	23.7	22.8	25.0	26.1	26.5	27.1	27.2	27.0	26.8
Break Even at Expiry	2.7	9.7	23.3	27.3	30.3	43.2	48.2	62.8	74.7	84.1	93.7	101.5	107.6	112.6
BPV per Contract	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00
Cal/Bus Days To Expiry	2/2	30/22	65/47	93/67	121/87	156/112	212/152	303/217	394/282	485/347	576/412	674/482	765/547	856/612
Futures Price	96-06	96-06	96-01+	96-01+	96-01+	95-30	95-30	95-28+	95-29	95-30	95-31	95-31	95-30+	95-30+

VOLATILITY is price volatility for bond, note and index options; yield volatility for money market options

Volatility Cones are based on number of days to expiration. For example, if 40 days remain, we look at actual volatility of the nearby contract in every 40 day period over the last 2 years

% Historical Above Implied shows how often actual volatility exceeded current implied volatility. A higher number means cheaper options.

BASIS POINT VOLATILITY is basis point yield volatility for all options.

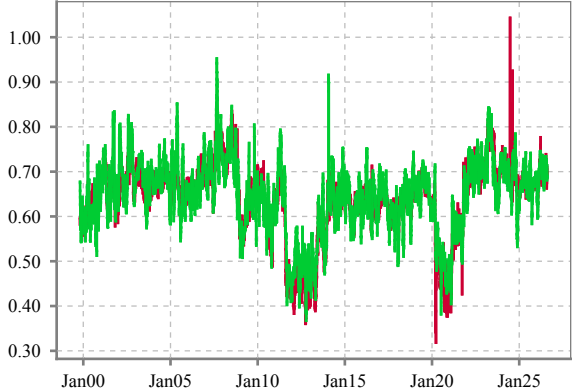
Yield Change Required to... is based on an at-the-money straddle. Yield change for bond, note and money market options; Price change for equity and commodity options

* Historical volatility is taken from old otr bond This

Derivatives Strategy

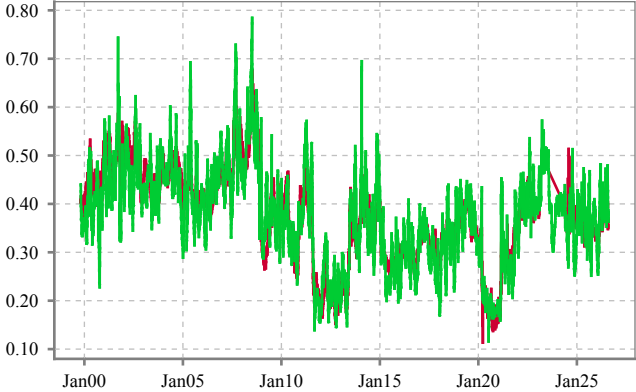
Maturity Structure of Treasury Volatility

Ratio of Price Volatility
(Front 5yr)/(Front 10yr)



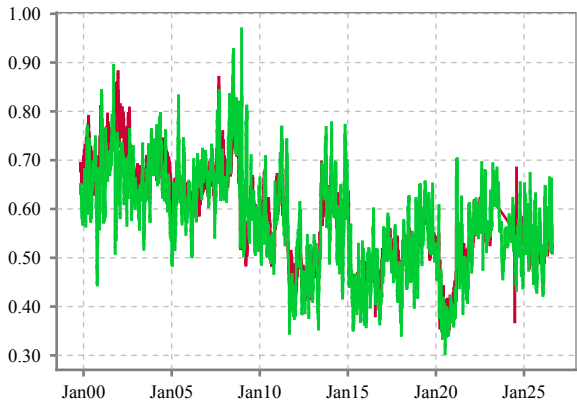
	Current	1 Yr Avg
Implied	0.71	0.69
Historical		0.69

Ratio of Price Volatility
(Front 5yr)/(Front 30yr)



	Current	1 Yr Avg
Implied	0.36	0.36
Historical		0.37

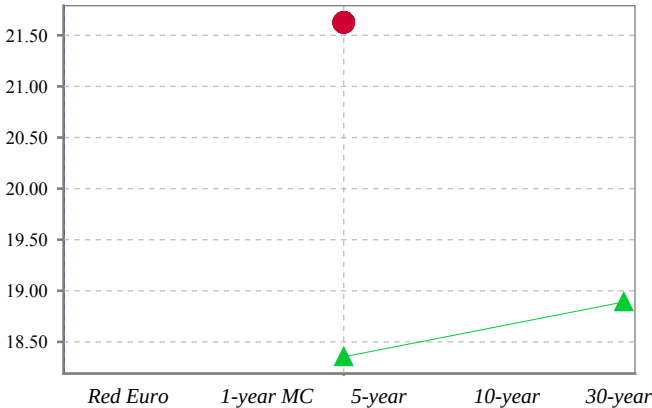
Ratio of Price Volatility
(Front 10yr)/(Front 30yr)



	Current	1 Yr Avg
Implied	0.51	0.52
Historical		0.54

— Implied
— 20-day Historical

Maturity Structure of Yield Volatility
Volatility



— Implied
— 20-day Historical

1.Contract switches 20 business days before quarterly option expiration.